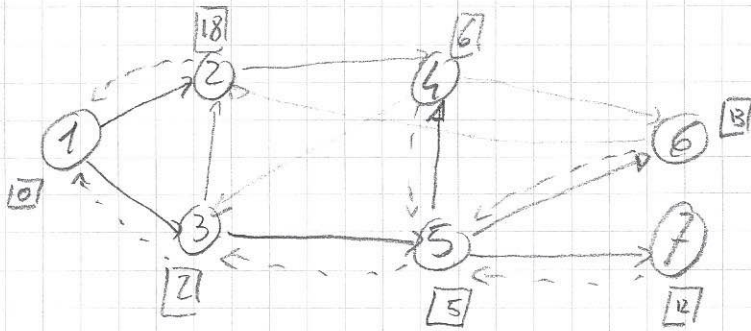


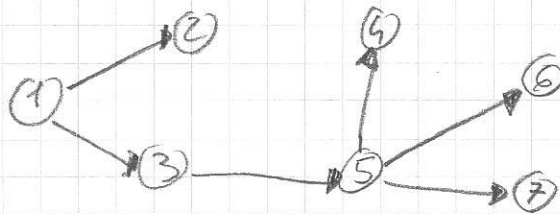
- 1) The graph contains cycles but the arc lengths are ≥ 0
 The most efficient algorithm in this case is SPT-Dijkstra



$$Q = \{2, 3, 5, 4, 6, 7\}$$

label 18 2 5 6 12 18

final SPT



$$\pi_6 - \pi_5 = 7$$

if $c_{56} \geq 7$ the solution does not change

$$2) B_1 = \{1, 2\}$$

$$A_{B_1} = \begin{bmatrix} -1 & 1 \\ -2 & -1 \end{bmatrix}$$

$$A_{B_1}^{-1} = \begin{bmatrix} -1/3 & -1/3 \\ 2/3 & -1/3 \end{bmatrix}$$

$$A_{B_1}^{-1} b_{B_1} = \begin{bmatrix} -2 \\ 2 \end{bmatrix}$$

$$A_3 \begin{bmatrix} -2 \\ 2 \end{bmatrix} = 4 + 2 = 6 \not\leq 4 \text{ violated}$$

B_1 is not primal feasible

$$c A_{B_1}^{-1} = [1, 1] \geq 0 \text{ dual feasible}$$

$$B_2 = \{2, 3\}$$

$$A_{B_2} = \begin{bmatrix} -2 & -1 \\ -2 & 1 \end{bmatrix}$$

$$A_{B_2}^{-1} = \begin{bmatrix} -1/4 & -1/4 \\ -1/2 & 1/2 \end{bmatrix}$$

$$A_{B_2}^{-1} b_{B_2} = \begin{bmatrix} -3/2 \\ 1 \end{bmatrix}$$

$$A_1 \begin{bmatrix} -3/2 \\ 1 \end{bmatrix} = 3/2 + 1 = 5/2 \leq 4$$

OK

primal feasible

$$c A_{B_2}^{-1} = [3/4, 3/4] \geq 0 \text{ dual feasible}$$

B_2 is an optimal base

3) Fixing

$x_6 = 0$ since it has a negative value and a positive weight

$x_7 = 1$ since it has a positive value and a negative weight

the problem becomes

$$P_0: \max 12x_1 + 7x_2 + 3x_3 + 3x_4 + 2x_5$$

$$9x_1 + 5x_2 + 2x_3 + 3x_4 + 3x_5 \leq 11$$

$$x_i \in \{0, 1\} \quad i=1 \dots 5$$

Ratio value_i / weight_i

$$\frac{12}{9} = 1.33 \quad \frac{7}{5} = 1.4 \quad \frac{3}{2} = 1.5 \quad \frac{3}{3} = 1 \quad \frac{2}{3} = 0.67$$

greedy $x_3 = 1 \quad 11 \rightarrow 9 \quad x_2 = 1 \quad 9 \rightarrow 4 \quad x_1 = 0 \quad x_4 = 1 \quad 4 \rightarrow 1 \quad x_5 = 0$

value = $3 + 7 + 3 = 13$ incumbent sol.

LP relaxation P_0

$$x_3 = 1 \quad 11 \rightarrow 9 \quad x_2 = 1 \quad 9 \rightarrow 4 \quad x_1 = \frac{4}{9} \quad ub = 3 + 7 + 12 \cdot \frac{4}{9} = 15 + \frac{1}{3}$$

$ub > \text{incumbent} \Rightarrow \text{Branch on } x_1$

$$P_1: 12 + \max 7x_2 + 3x_3 + 3x_4 + 2x_5$$

$$5x_2 + 2x_3 + 3x_4 + 3x_5 \leq 2$$

$$x_i \in \{0, 1\} \quad i=2 \dots 5$$

fixing $x_2 = x_4 = x_5 = 0$ since their weight > 2

$$12 + \max 3x_3$$

$$2x_3 \leq 2$$

$$x_3 \in \{0, 1\}$$

LP relaxation of P_1 $x_3 = 1$ optimal value 15 new incumbent

$$P_2: \max 7x_2 + 3x_3 + 3x_4 + 2x_5$$

$$5x_2 + 2x_3 + 3x_4 + 3x_5 \leq 11$$

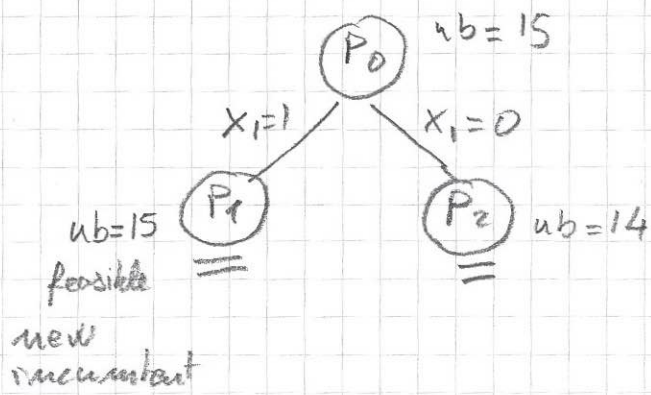
$$x_i \in \{0, 1\} \quad i=2 \dots 5$$

LP relaxation of P_2

$$x_3 = 1 \quad 11 \rightarrow 9 \quad x_2 = 1 \quad 9 \rightarrow 4 \quad x_4 = 1 \quad 4 \rightarrow 1 \quad x_5 = 1/2 \quad ub = 3 + 7 + 3 + 1 = 14$$

$ub \leq \text{incumbent} \rightarrow \text{STOP}$

optimal solution $\therefore x_1 = 1 \quad x_3 = 1 \quad \text{value} = 15$



incumbent 13

4) $\min \quad cx$
 $Ax \leq b$

$\max \quad yb$
 $yA = c$
 $y \leq 0$

$\bar{x}$ feasible

$$A\bar{x} \leq b$$

multiply on the left by $\bar{y} \leq 0$

$$\bar{y} A \bar{x} \geq \bar{y} b$$

the inequality is $\geq$ since we multiplied by $\bar{y} \leq 0$

$$\bar{y} \text{ feasible} \Rightarrow \bar{y} A = c$$

$$c \bar{x} \geq \bar{y} b$$

q.e.d

5) Sets $D = \{1 \dots T\}$ days

Parameters

Q normal capacity

Q^+ additional capacity

m_0 initial quantity

q minimum quantity

d_t demand of the market $t \in D$

P_t purchase price $t \in D$

r_t risk cost for going below the minimum $t \in D$

c_t cost of the additional capacity $t \in D$

Variables

$x_t \geq 0$ purchased quantity in day t $t \in D$

$y_t \in \{0, 1\}$ access to the extra capacity $t \in D$

$z_t \geq 0$ quantity in the depot $t \in D \cup \{0\}$

$\alpha_t \geq 0$ quantity below the minimum $t \in D$

Multiperiod constraints

$$z_t = z_{t-1} + x_t - d_t \quad t \in D$$

$$z_0 = m_0$$

Upper capacity constraints

$$z_t \leq Q + Q^+ y_t \quad t \in D$$

Lower capacity constraints

$$z_t \geq q - \alpha_t$$

Objective function

$$\min \sum_{t \in D} P_t x_t + \sum_{t \in D} c_t y_t + \sum_{t \in D} r_t \alpha_t$$

Variant

new multiperiod constraint

$$z_t = z_{t-1} + x_{t-1} + d_t \quad t \in D$$

6)

 $N = \{1 \dots n\}$ days

 $A = (h, k) \rightarrow \text{reservation } i \quad i = 1 \dots m$
 $h = s_i \quad k = e_i$
 $c_{hk} = -e_i \quad u_{hk} = 1$
 $(h, h+1) \rightarrow \text{empty} \quad h = 1 \dots m-1$
 $c_{hk} = 0 \quad u_{hk} = m$
 $(m, 1) \rightarrow \text{circulation arc}$
 $c_{m1} = 0 \quad u_{m1} = m$

Balance of the nodes = 0

